

AWS Data Engineer

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PROFESSIONAL SUMMARY:

- Experienced AWS Data Engineer with over 7+Years of hands-on expertise in Big Data and Machine Learning.
- Extensive experience in handling large data sets of both structured and unstructured data. Proficient in Data Acquisition, Data Validation, Predictive Modeling, Statistical Modeling, Data Modeling, and Data Visualization.
- Specialized in Data Warehousing and Decision Support Systems and extensive experience in implementing Full Life cycle Data Warehousing Projects and in Hadoop/big data-related technology experience in Storage, Querying, Processing, and analysis of data.
- Proven expertise in ETL development, including scripting ETL pipelines in Python for ingesting files from AWS S3 to Redshift and in developing robust ETL processes for data extraction, transformation, and loading from various sources into data warehouses.
- Demonstrated skill in designing and optimizing data warehouses to ensure optimal performance, storage, and retrieval of data.
- Skilled in utilizing Databricks for efficient data processing and pipeline management, including optimizing Spark jobs for performance and scalability.
- Experienced in implementing end-to-end data solutions on Databricks, integrating with other cloud services for seamless data engineering workflows.
- Proficient in all phases of the data lifecycle with a focus on crafting robust data warehousing and integration solutions utilizing Informatica PowerCenter.
- Proficient in designing and implementing Snowflake schema modeling for complex analytical queries, ensuring efficient data storage and retrieval in the data warehouse design.
- Expertise in implementing Service-Oriented Architectures (SOA) with XML-based Web Services (SOAP/REST) for seamless integration and communication between systems.
- Developed Python scripts to parse XML, and JSON files and loaded the data in Snowflake Data warehouse.
- Strong programming skills in Python, R and scala for building robust data solutions, statistical modeling, and automating data workflows.
- Experience in utilizing Python libraries such as NumPy, Pandas, and Scikit-learn for data analysis and model building.
- Good understanding of Statistics and developing Machine learning models such as Linear Regression, Logistic Regression, Decision Trees, RF, SVM.
- Hands-on experience in Hadoop Eco - system components like HDFS, Cloudera, YARN, Hive, HBase, Sqoop, Flume, Kafka, Impala, Airflow and Programming in Spark using Python and Scala.
- Proficient in using tools like Sqoop and Flume for directing data to/from HDFS, and experience with Oozie and Zookeeper for workflow scheduling and coordination.
- Experienced in implementing data models and loading unstructured data with HBase, DynamoDB, and Cassandra. Proficient in Spark architecture and its integrations (Spark SQL, DataFrames, Datasets APIs).
- Adept at migrating on-premises Hadoop platforms to AWS cloud solutions and skilled in using R for statistical procedures, including ggplot, shiny apps, and dashboard building.
- Proficient in Integration Services (SSIS), Reporting Service (SSRS), and Analysis Services (SSAS) for comprehensive BI solutions.
- Knowledge of database design concepts, including normalization, indexing, and data partitioning, as well as experience with database management systems such as MySQL, PostgreSQL, and Oracle.
- Proficient in SQL and NoSQL databases including Snowflake, MongoDB, Oracle, and MySQL for comprehensive data management. Skilled in database management, data modeling, schema design, and optimizing performance in both relational and non-relational database environments.
- Skilled in utilizing AWS Stack (S3, EMR, EC2, SQS, Glue and RedShift, IAM, CloudWatch, EKS, EBS, CloudFront) for efficient data management, processing and analytics.
- Experienced in working with AWS cloud and EMR clusters with Spark for raw data processing and accessing data from S3 buckets.
- Experienced in using Terraform for automating the deployment and management of infrastructure as code in AWS environments.
- Proficient in creating data pipelines, workflow scheduling and monitoring experience using Apache Airflow.

- Skilled in utilizing AWS Glue for efficient ETL processes and data cataloging. Experience in using Amazon Redshift for data warehousing and large-scale data storage solutions.
- Developed and optimized Apache Spark jobs using PySpark and Scala for faster data processing in a test environment, utilizing Spark Core, RDDs, DataFrames, Spark Streaming, and Spark SQL for querying and computational analytics
- Implemented a POC to migrate MapReduce jobs to Spark, integrated Spark with Hadoop, Hive, and SQL/Oracle, and leveraged Spark MLlib for machine learning tasks.
- Experience in using Apache Kafka for collecting, aggregating, and moving large amounts of data.
- Well-versed in version control and CI/CD tools such as GIT, SourceTree, Bitbucket, and Jenkins for automated build and deployment processes.
- Experience working with CI/CD pipelines using tools like Jenkins, Maven/Ant, and Git for continuous integration and delivery.
- Skilled in version control tools like Git and Bitbucket, experienced in methodologies such as Agile and Waterfall, and proficient in unit testing and frameworks for software development.
- Flexible working with Operating Systems like Unix/Linux (Centos, Red hat, Ubuntu) and Windows Environments.
- Proficient in migrating data between database systems and cloud platforms, implementing data quality measures, and ensuring data integrity, security, and regulatory compliance across the data lifecycle.
- Proficient in Agile Methodologies, with involvement in designing and developing data pipeline processes for various modules within AWS.

TECHNICAL SKILLS:

Programming Languages	Python (Jupyter, PyCharm IDE), R, Java, Scala, SQL, PL/SQL, SAS
Scripting Languages	Bash, Python, R Language, YAML, Shell Scripts
ETL	PySpark, CTRL-M, Data Stage, Informatica Power Center, Talend, Pentaho, Microsoft SSIS, DataStage 7.5, Ab Initio, MS Fabric
Big Data Tech	Hadoop, Spark, MapReduce, YARN, Hive, Pig, Sqoop, Oozie, Maven
Data Ingestion	Sqoop, Flume, Kafka
Cloud Computing Tools	Snowflake, AWS, Databricks, Amazon EC2, EMR, S3
NoSQL Databases	HBase, Cassandra, MongoDB, Apache, Hadoop HBase
Hadoop Ecosystem	HDFS, MapReduce, Yarn, Oozie, Zookeeper, Cloudera
Operating Systems	Unix, Linux, Mac OS, Windows, Ubuntu
Databases	Snowflake Cloud DB, AWS Redshift, AWS Athena, Oracle, MySQL, Teradata 12/14, DB2 10.5, MS Access, SQL Server, PostgreSQL 9.3, Netezza, AmazonRDS
SQL Server Tools	SQL Server Management studio, Enterprise Manager, Query Analyzer, Profiler, Export, and Import (DTS)
IDE	IntelliJ, Eclipse, Visual Studio, IDLE
Web Services	Restful, SOAP, O9iAS, Oracle Form Server, WebLogic 8.1/10.3, Web Sphere MQ 6.0
Methodologies	Agile, Waterfall
Data Analysis	Data mining, Predictive modeling, Statistical modeling, Data modeling, Data Visualization.
Reporting / BI Tools	MS Excel, Tableau, Tableau Server and Reader, Power BI, QlikView, Crystal Reports, SSRS, Splunk

PROFESSIONAL EXPERIENCE:

PennyMac, Westlake Village, CA

March 2022- Till

Present

AWS Data Engineer

Responsibilities:

- Designed, developed, and deployed comprehensive data pipelines and analytics solutions using AWS, Databricks, Snowflake, and other cutting-edge technologies.
- Implemented AWS-based ETL pipelines using EC2, S3, RDS, Lambda, and Glue, optimizing data ingestion and transformation processes.
- Implemented data validation and cleansing routines within SnapLogic pipelines to ensure data quality.
- Expertise in various Relational databases including MySQL, Oracle, and MS SQL Server, as well as NoSQL databases such as MongoDB, HBase, and Cassandra.
- Utilized Azure Databricks to build and manage data pipelines, leveraging Apache Spark for large-scale data processing.
- Designed and implemented data workflows using Prefect, orchestrating complex ETL pipelines with seamless scheduling, monitoring, and error handling.
- Developed and optimized SQL queries, stored procedures, and views in DB2 to support data extraction, transformation, and loading (ETL) processes for high-volume dataset.
- Participated in building reports and team meetings to explain the benefits of various data lake and data warehouse designs, utilizing tools like Amazon Redshift, Athena, and DynamoDB.
- Implemented data pipelines in Microsoft Fabric using Data Factory for orchestrating data workflows and automating ETL processes.
- Developed Databricks clusters for both batch and streaming data processing, employing PySpark jobs and performing data transformations.
- Designed and implemented dimensional data models and star schemas in Snowflake to support financial modeling and data analysis.
- Built and managed data pipelines in Snowflake and DBT, employing SnowSQL and SnowPipe for data manipulation and automation.
- Migrated Oozie workflows to Airflow to automate data pipelines and streamline ETL operations.
- Leveraged Amazon Athena to analyze large datasets in S3 with SQL queries.
- Developed and maintained Apache Airflow DAGs to schedule and orchestrate ETL processes, automating data extraction from weblogs and other sources.
- Collaborated with business stakeholders to define KPIs and build interactive dashboards for data visualization using Tableau and Power BI.
- Built SnapLogic pipelines to ingest data from Oracle, MongoDB, and SFTP, managing server tasks and operations.
- Worked closely with data scientists and analysts to provide them with clean, structured data for analytics and machine learning.
- Implemented Azure Stream Analytics for real-time data processing and monitoring of event-driven architectures.
- Designed machine learning models in AWS SageMaker to deliver predictive analytics and deeper insights.
- Proficient in Apache Spark, EMR, AWS Glue, and other cloud-based serverless compute technologies for large-scale data processing and analytics.
- Provided technical expertise and solution recommendations during pre-sales engagements, aligning data engineering solutions with client requirements.
- Experience working on Java to implement complex data processing algorithms and workflows for batch and real-time data integration.
- Monitored and optimized SnapLogic jobs for performance, ensuring minimal downtime and maximizing throughput.
- Utilized DataStage's connectors and stages to manage data flow between on-premise and cloud environments, ensuring smooth data migration.
- Implemented data governance frameworks to maintain data quality and integrity across distributed databases, such as Cassandra and MongoDB.
- Automated testing and validation of data pipelines to ensure data integrity and reliability during the testing phase of SDLC.
- Supported proposal development by drafting the technical approach, outlining data transformation strategies, and creating cost-effective solutions for clients.
- Utilized Git and Bitbucket for version control, with Maven for build automation, following best practices in software development and unit testing.

- Collaborated with cross-functional teams to gather requirements and define data processing needs within the SDLC framework.
- Experienced in Agile/DevSecOps environments, leveraging iterative development and automation for continuous delivery.
- Collaborated with stakeholders to design tailored Big Data and cloud-based solutions, demonstrating the capabilities of AWS services like S3, EMR, Redshift, and Glue.
- Maintained data warehouses using DataStage, facilitating business intelligence and analytics initiatives.
- Designed and deployed RESTful and SOAP web services for data interchange between distributed systems.
- Designed and implemented scalable data pipelines using SnapLogic for seamless data integration across various systems and applications.
- Utilize Terraform and Azure Resource Manager (ARM) templates to automate infrastructure deployment and resources.
- Skilled in architecting and enhancing data platforms as well as implementing service-oriented architectures to support enterprise data needs.
- Performed data cleansing, transformation, and enrichment tasks using PySpark, ensuring high-quality data for downstream analytics and machine learning models.
- Collaborated with cross-functional teams to deploy scalable data solutions and optimize performance in Microsoft Fabric environments.
- Automated data ingestion and transformation processes by developing workflows using Apache Airflow, reducing manual intervention and enhancing workflow reliability.
- Designed and implemented data transformation workflows using DBT, enhancing data pipeline efficiency and data quality across ETL processes.
- Implemented efficient indexing and partitioning strategies within DB2 to enhance performance and reduce query run time for complex analytical queries.
- Managed Kubernetes resources, including Pods, Deployments, Services, and ConfigMaps, for containerized applications.
- Performed data validation using SQL and UNIX commands for backend testing, ensuring data accuracy and reliability.

**First American Bank, Elk Grove Village, IL
2022**

June 2019- Feb

AWS Data Engineer

Responsibilities:

- Worked as a Data Engineer within an Agile environment with two-week sprint cycles, effectively dividing and organizing tasks.
- Created ETL packages to load data from Excel files, RDBMS tables, and other sources into staging tables, incorporating various transformations.
- Designed and deployed machine learning models in AWS SageMaker for data pre-processing, model training, and predictive analytics.
- Collaborated with data science teams to manage dependencies and trigger machine learning models streamlining data-to-insight workflows.
- Implemented complex data workflows in DataStage to process structured and unstructured data from multiple sources.
- Built and optimized back-end data processing systems using Java, focusing on performance, scalability, and reliability.
- Managed Databricks clusters for both batch and streaming data processing, optimizing PySpark jobs for data transformation and analysis.
- Utilized DBT's testing framework to build robust data quality checks, ensuring the accuracy and consistency of data in production environments
- Employed Spark Streaming in Databricks for real-time data ingestion and analytics.
- Utilized HDFS, HBase, SQS, SNS, and MapReduce for processing large-scale datasets.
- Designed and implemented dimensional data models and star schemas in Snowflake to facilitate financial modeling and analytics.
- Integrated DB2 with ETL workflows using tools like Apache Airflow and AWS Data Pipeline, facilitating seamless data migration and transformation across different platforms.
- Built and managed data pipelines in Snowflake and DBT, utilizing Snow SQL and Snow Pipe for efficient data manipulation and automation.

- Applied best practices in data modeling for scalability, flexibility, and future growth in high-velocity data environments.
- Conducted performance tuning of DataStage jobs by optimizing parallelism, partitioning, and memory management to handle large volumes of data.
- Developed RESTful APIs and Web Services using Java and Spring Boot to enable real-time data processing and retrieval.
- Leveraged Amazon Athena for SQL-based analytics on large datasets stored in S3.
- Ensured data governance, security compliance, and cost optimization across AWS accounts using AWS CloudTrail.
- Developed and maintained Apache Airflow DAGs to orchestrate and automate ETL processes, including data extraction from weblogs.
- Collaborated with business stakeholders to define KPIs and build interactive dashboards using Tableau and Power BI for data visualization and reporting.
- Developed SnapLogic pipelines to ingest data from sources like Oracle, MongoDB, and SFTP, while managing server tasks and operations.
- Optimized PostgreSQL databases by designing efficient schemas and writing complex SQL queries for ETL workflows.
- Optimized DBT transformations for performance, reducing query runtime and improving overall efficiency in data warehousing environments.
- Collaborated with DBAs to design and implement DB2 database schemas for data warehousing, ensuring scalability and reliability for analytics applications.
- Managed semi-structured data ingestion using MongoDB, combined with relational databases for structured data storage.
- Designed and deployed containerized microservices using Docker and Kubernetes, enabling modular and scalable data processing.
- Integrated PySpark with Apache Airflow for building and scheduling data workflows, streamlining data pipeline automation.
- Integrated data from heterogeneous sources including databases (Oracle, SQL Server, DB2), flat files, and cloud platforms using DataStage.
- Designed and developed batch processing and real-time processing solutions using ADF, Databricks clusters, and stream analytics.
- Managed Kubernetes resources, including Pods, Deployments, Services, and ConfigMaps, for containerized application environments.
- Implemented CI/CD pipelines with Git, Bitbucket, Maven, and other tools, ensuring streamlined version control and automated deployments.

Electronic Arts, Austin, TX
2019

April 2017-May

Big Data Engineer

Responsibilities:

- Analyzed both functional and non-functional requirements to architect scalable data solutions, installing and configuring Hadoop MapReduce and HDFS for robust large-scale data processing infrastructure.
- Developed complex MapReduce jobs in Java for data cleaning and preprocessing, optimizing performance and data accuracy.
- Created and managed Pig scripts in Pig Latin to streamline data transformation workflows, enhancing processing efficiency.
- Automated data processing tasks by developing MapReduce jobs using Pig Latin, significantly improving data ingestion and system performance.
- Employed Sqoop for seamless data transfer between MySQL and HDFS, ensuring consistent data integration and flow.
- Wrote optimized SQL queries for data manipulation in MySQL and Oracle, improving data retrieval and processing performance.
- Crafted HiveQL queries for comprehensive data analysis, creating and managing Hive tables to support in-depth data exploration.
- Executed Hive queries in a MapReduce environment, integrated with HDFS to enable large-scale data operations.
- Developed a custom filesystem plugin for Hadoop to enhance data accessibility across Hadoop MapReduce, HBase, Pig, and Hive, enabling seamless data platform operations.
- Designed and implemented a large-scale parallel relation-learning system based on MapReduce, supporting advanced data processing.

- Configured and benchmarked Hadoop clusters for internal use, optimizing performance, reliability, and troubleshooting system issues through log file management.
- Designed and implemented batch jobs and scheduling scripts to automate Hadoop program execution, reducing manual intervention and optimizing data processing workflows.
- Automated data collection and transformation tasks by developing Python scripts for extracting and processing data from online news websites, ensuring efficient and timely data processing.
- Applied Python scripting to enhance data acquisition workflows, automating data collection and preprocessing tasks from various online sources.
- Conducted performance tuning and query optimization for Oracle and MySQL databases, employing advanced indexing strategies, partitioning, and normalization techniques to improve data retrieval efficiency and reduce query execution times.
- Developed and managed ETL pipelines using Sqoop to import data from MySQL and PostgreSQL databases into HDFS, utilizing SQL and PL/SQL for seamless data transformation, migration, and integration across diverse data platforms.

EDUCATION:

Masters in computer science – University of Maryland Baltimore County

Bachelors in Electronics and communication –Visvesvaraya national institute of technology